COVID-19 Chest X-Ray Analysis and Survival Prediction

Machine Learning Baselines Using Pretrained Deep Learning Features

1. Problem Statement

The objective of this project is to investigate whether features extracted from chest X-ray images can help predict patient survival outcomes and evaluate how well these predictions generalize across geographical regions. Medical imaging datasets may contain differences in patient populations, imaging practices, and disease prevalence across countries. Consequently, a model that performs well on one population may not generalize effectively to another.

This project addresses the problem by extracting representations from chest X-ray images using a pretrained deep learning model and evaluating multiple machine learning baselines for survival prediction. Rather than training a convolutional neural network from scratch, the approach leverages pretrained representations to investigate their usefulness for a downstream clinical prediction task.

A key objective is to assess geographical generalization. The evaluation therefore holds out geographical regions during testing, allowing the experiments to investigate whether models trained on patients from other regions can distinguish between patients who survived and those who did not in an unseen region.

> **Primary task:** Binary classification of patient survival using chest X-ray-derived features, evaluated through leave-one-region-out validation.

2. Dataset Details

The project uses the publicly available COVID-19 chest X-ray dataset provided through the COVID-19 Chest X-ray Dataset repository. The dataset contains chest radiographs and associated metadata describing patient characteristics, clinical findings, imaging views, geographical location, and clinical outcomes.

The implementation loads the images and metadata using the `COVID19_Dataset` interface from TorchXRayVision. It restricts the image views to anteroposterior (AP) and posteroanterior (PA) radiographs. Additional metadata processing associates images with clinical labels and geographical information.

Dataset characteristics

- **Input:** Chest X-ray images in AP or PA projection.
- **Metadata:** Clinical findings, survival status, patient information, image view, and geographical location.
- **Target variable:** Survival status, represented as a binary label indicating survival or non-survival.
- **Geographical groups:** Regions are used to separate training and evaluation data.
- **Eligibility filtering:** The survival task includes records with known survival status, excludes records marked as having intubation present, and restricts the recorded image offset to greater than zero and less than eight days.

The filtering procedure is important because the model should be evaluated only on records satisfying the defined task criteria. The final sample count depends on the available metadata, missing values, and the application of these filters. The experiments use the eligible records rather than assuming that every image in the original repository has a valid survival label.

3. Approach

The project follows a feature-extraction and downstream-classification pipeline. First, chest X-ray images are preprocessed to provide standardized inputs to a pretrained DenseNet model. Next, image representations are extracted and used as inputs to machine learning classifiers. Finally, the models are evaluated using geographical holdouts.

3.1 Image preprocessing and feature extraction

TorchXRayVision is used to load a pretrained DenseNet with its available pretrained weights. The images undergo center cropping and resizing to 224 × 224 pixels. The network is then used to obtain intermediate image features and predictions for a set of chest radiographic pathologies. The extracted features are reused for the downstream prediction task, avoiding the need to train the entire image network from scratch.

3.2 Baseline comparison

The original experiments compare several representations, including intermediate DenseNet features, the model's pathology outputs, selected pathology outputs, and a baseline based on the prevalence of the target class. These comparisons help determine whether image-derived representations provide predictive information beyond a simple baseline.

4. Implementation Overview

The implementation is developed in Python using Jupyter Notebook. PyTorch and TorchXRayVision handle image loading, preprocessing, and feature extraction, while Pandas and NumPy support metadata processing and numerical operations. Scikit-learn provides the downstream classification algorithms and evaluation metrics.

4.1 Feature extraction and label preparation

The notebook processes the dataset in batches using a PyTorch DataLoader. During feature extraction, gradient computation is disabled because the pretrained network is used for inference rather than backpropagation. Intermediate feature vectors, pathology predictions, and image data are collected for subsequent analysis.

The notebook constructs the survival target from the recorded survival metadata. It applies the eligibility filters described above and assigns each eligible example to a geographical region. Records with unknown geographical regions are excluded from the additional region-aware classifier benchmark.

4.2 Geographical validation

The evaluation uses a leave-one-region-out strategy. In each fold, one geographical region is held out for testing while the remaining eligible regions provide training data. This approach tests cross-region generalization rather than relying exclusively on a random split of individual examples.

The notebook also includes an extended benchmark using four classifiers: logistic regression, a radial-basis-function support vector classifier, Extra Trees, and Random Forest. Logistic regression and the support vector classifier use standardized features. The tree-based models use 500 estimators, and class weighting is enabled to help account for unequal class frequencies.

Performance is evaluated using the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC). AUROC measures ranking discrimination across thresholds, while AUPRC summarizes the precision-recall trade-off and is particularly informative when the positive class is more prevalent.

5. Results and Discussion

In the additional region-aware classifier benchmark, Extra Trees achieved the strongest mean AUROC among the four evaluated models. The results recorded in the notebook are summarized below.

Classifier	Mean AUROC	Mean AUPRC
Extra Trees	0.697	0.943
Random Forest	0.683	0.935
Logistic Regression	0.678	0.936
RBF Support Vector Classifier	0.564	0.895

Extra Trees obtained a mean AUROC of 0.697 and a mean AUPRC of 0.943, outperforming the other classifiers on both reported metrics. Random Forest and logistic regression produced similar AUROC values, while the RBF support vector classifier achieved the lowest AUROC in this benchmark.

The comparatively high AUPRC values should be interpreted alongside the class distribution. Precision-recall performance can remain high when the positive class is common, even when the model's ability to

rank positive and negative examples is only moderate. Therefore, the AUROC results suggest that survival discrimination remains challenging across geographical regions.

These results are exploratory rather than evidence of clinical readiness. The held-out regions and eligible patient counts are limited, and performance may vary across regions. The experiment also evaluates image-derived features alongside clinical outcome metadata; it does not establish that chest X-rays alone are sufficient for reliable survival prediction.

6. Conclusion

This project implements a machine learning pipeline for investigating COVID-19 patient survival prediction from chest X-ray images. A pretrained DenseNet is used to extract image representations, which are then evaluated with conventional classifiers under a geographical holdout strategy. This design provides a practical baseline for studying the transferability of pretrained medical imaging features across different patient populations.

Among the classifiers in the extended benchmark, Extra Trees achieved the best reported performance, with an AUROC of 0.697 and an AUPRC of 0.943. However, the moderate AUROC indicates that meaningful cross-region discrimination remains difficult. Future work should include larger and more representative datasets, patient-level deduplication, more extensive validation, uncertainty estimation, and careful assessment of demographic and geographical biases. Independent external validation would also be necessary before considering any clinical application.

End of Report